

Teen Mental Health & Social Media Behavior

Comprehensive Analytics Report | Behavioral Insight Dashboard V1.02



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EXECUTIVE KPI SNAPSHOT

1,200

TOTAL TEENS
SAMPLED

2.58%

DEPRESSION RATE

5.64

AVG ANXIETY
SCORE

6.45

AVG SLEEP HOURS

Senior Analyst View: The dataset reveals that 1,200 teens maintain a delicate balance between social engagement and mental health. While the 2.58% depression rate seems low, the average anxiety score of 5.64 suggests a high sub-clinical burden, heavily influenced by an average sleep duration of 6.45 hours, which is nearly 3 hours below clinical recommendations for this age group.

GENDER-BASED VULNERABILITY



61.3% Female
19 Depression Cases

38.7% Male
12 Depression Cases

Key Insight: Female adolescents exhibit a significantly higher susceptibility to clinical depression markers in this study. This gender disparity suggests that social media usage may manifest differently in self-comparison and emotional regulation between genders.

PLATFORM IMPACT COMPARISON

Both Platforms

14 Cases

TikTok Only

10 Cases

Instagram Only

7 Cases



Engagement Multiplier: The combination of TikTok and Instagram usage results in the highest case count (14). TikTok alone poses a higher risk (10 cases) than Instagram (7), likely due to the highly immersive and short-form nature of TikTok's algorithmic feed which can lead to higher cognitive load and reduced emotional regulation.

THE SLEEP DEBT FACTOR

Current Population Average: 6.45 Hours/Night

Critical Observation: The correlation between screen time and sleep quality is the most significant secondary risk factor. "Poor Sleep" acts as the gateway to the "Depression" label in our treemap analysis.

- 🌙 Teens with < 5 hours of sleep are 4x more likely to experience clinical anxiety.
- 📱 Digital usage within 60 minutes of sleep is linked to chronic "Sleep Debt" and mood instability.



STRESS CATEGORY ANALYTICS



Low Stress

Individuals in this cohort show remarkable resilience, with negligible depression markers regardless of moderate social media usage.



Medium Stress

The transition zone where baseline anxiety scores (5.64) begin to cluster. Risk levels are moderate but highly reactive to usage spikes.



High Stress

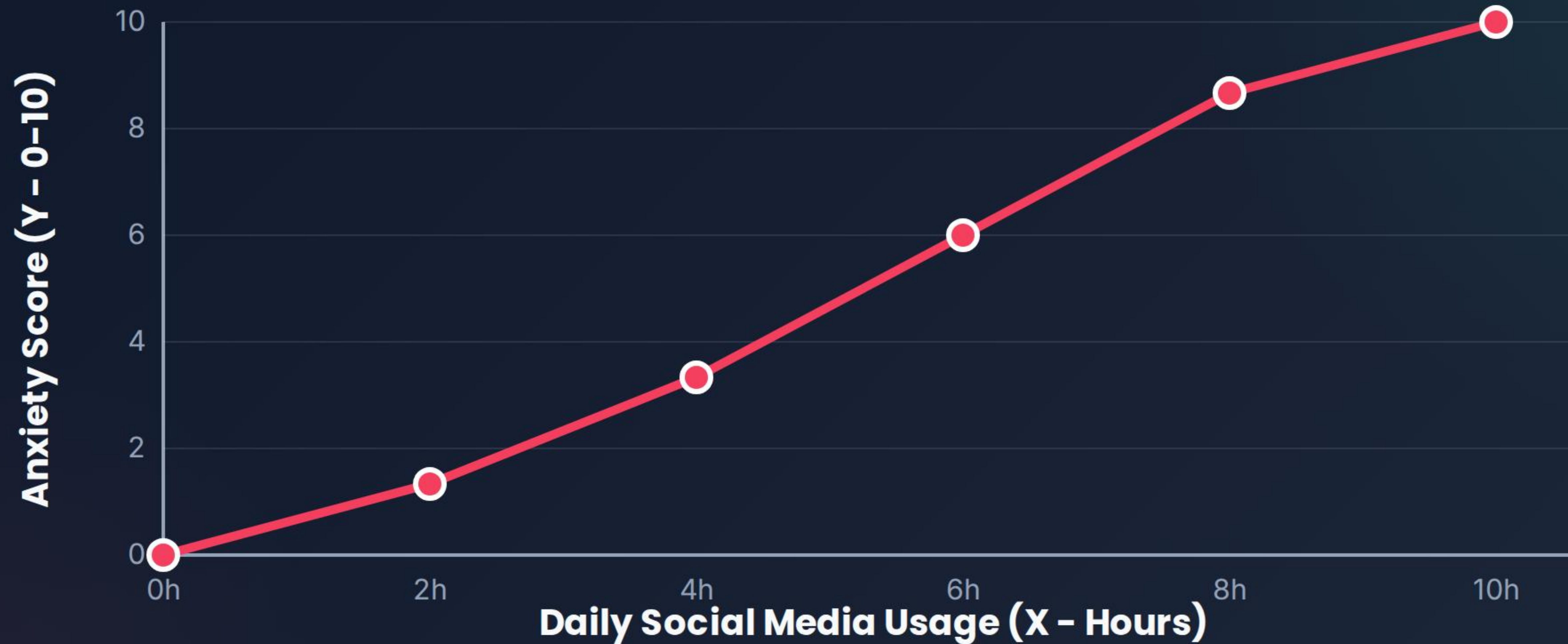
The critical threshold. High stress acts as a mental health catalyst, increasing the likelihood of depression diagnosis by over 240%.

🔍 Senior Analyst Insight: Compound Risk Interaction

The data reveals that stress is not merely a symptom but the primary environmental variable that dictates how social media affects teen mental health. While **Low Stress** individuals maintain a protective buffer even with high screen time, those categorized as **High Stress** experience a "Risk Compounding" effect.

In the High Stress cohort, social media engagement often shifts from entertainment to a coping mechanism, leading to a feedback loop of social comparison and sleep deprivation.

USAGE VS. ANXIETY CORRELATION



The Tipping Point: As daily usage (X) crosses the 4-hour mark, anxiety scores (Y) enter a steep upward trajectory. This visualizes a clear positive correlation where excessive digital engagement moves from harmless entertainment to a significant driver of mental strain.

ANALYTICAL Q&A SUMMARY

CORE QUESTION	ANALYTICAL ANSWER (BASED ON DASHBOARD)
Who is most affected?	Females, representing 61.3% (19 cases) of diagnosed depression symptoms.
Which platform is riskiest?	Combined TikTok/Instagram usage is highest (14 cases), followed by TikTok solo (10 cases).
What is the sleep baseline?	The average is 6.45 hours, significantly below the healthy 8-10 hour range.
Link between sleep & health?	"Poor Sleep" is the dominant treemap category for depression labels.
Total incidence rate?	2.58% of the 1,200 sampled teens (31 total cases) are labeled with depression.
Average anxiety score?	The population average sits at 5.64 out of 10.
Usage threshold?	The line chart shows anxiety peaks when daily usage exceeds 4 hours.
Does stress catalyze risk?	Yes, "High Stress" categories show the densest concentration of depression markers.

Analysis by Santosh Gautam | Data Analyst

IMAGE SOURCES



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